

MSDM5004

Numerical Methods and Modeling in Science
Spring 2026

Lecture 12

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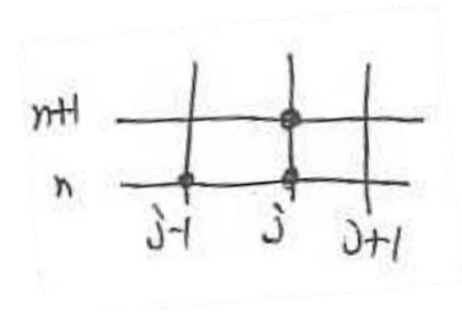
2.6. The CFL condition

Stability condition

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$$

Consider the numerical scheme

$$\frac{U_j^{n+1} - U_j^n}{\Delta t} + a \frac{U_j^n - U_{j-1}^n}{\Delta x} = 0 \quad (2)$$



$$U_j^{n+1} = U_j^n - a\nu (U_j^n - U_{j-1}^n)$$

$$\nu = \frac{\Delta t}{\Delta x}$$

$$U_j^{n+1} = (1 - a\nu)U_j^n + a\nu U_{j-1}^n$$

$$U_j^{n+1} = (1 - a\nu)U_j^n + a\nu U_{j-1}^n$$

U_j^{n+1} depends on U_j^n and U_{j-1}^n

depends on
 U_j^{n-1}, U_{j-1}^{n-1}

depends on
 $U_{j-1}^{n-1}, U_{j-2}^{n-1}$

$\Rightarrow U_j^{n+1}$ depends on $U_j^{n-1}, U_{j-1}^{n-1}, U_{j-2}^{n-1}$

Repeat this consideration, we can find that

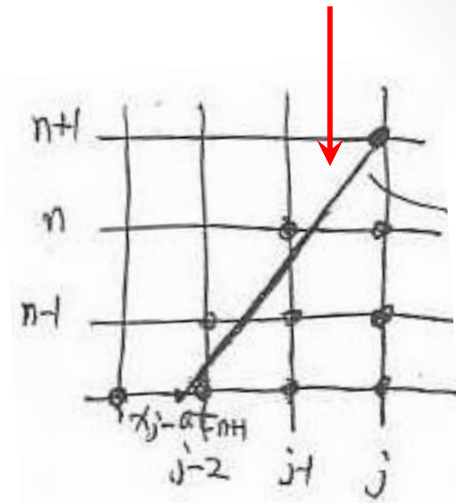
U_j^{n+1} depends on $U_j^0, U_{j-1}^0, U_{j-2}^0, \dots, U_{j-n-1}^0$.

Numerical domain of dependence of U_j^{n+1} is

$$\{x_{j-n-1}, x_{j-n}, \dots, x_{j-1}, x_j\}$$

which for small Δx is approximately the interval $x_{j-n-1} \leq x \leq x_j$.

characteristic:
 $x - at = \text{constant}$



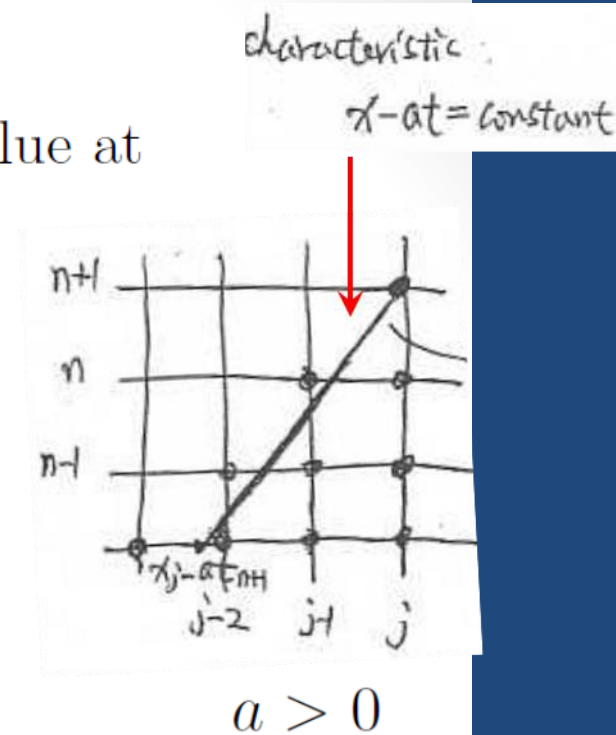
$$a > 0$$

The exact solution $u(x_j, t_{n+1})$ depends on the value at

$$(x_j - at_{n+1}, 0).$$

The domain of dependence of $u(x_j, t_{n+1})$ is

$$\{x_j - at_{n+1}\}.$$



A **necessary** condition for stability:

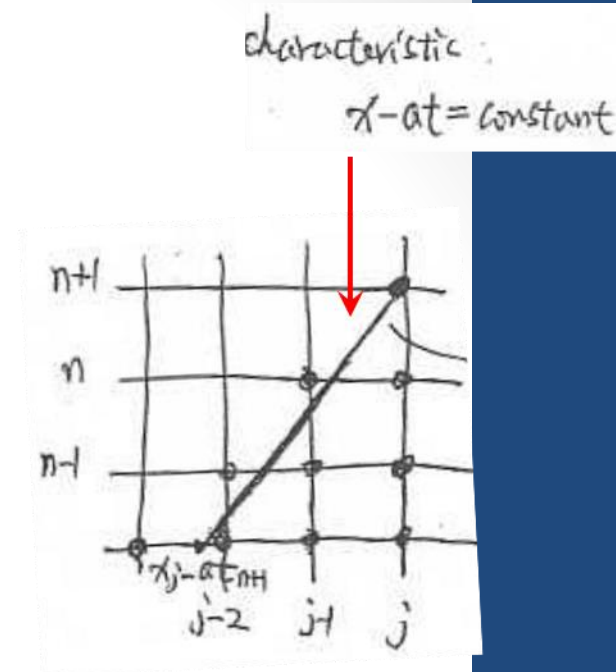
Domain of dependence is inside the numerical domain of dependence.

This is the CFL condition for stability.

CFL: Courant, Freidrichs, Lewy

The CFL condition of this scheme is $x_{j-n-1} \leq x_j - at_{n+1} \leq x_j$.

CFL condition $x_{j-n-1} \leq x_j - at_{n+1} \leq x_j.$

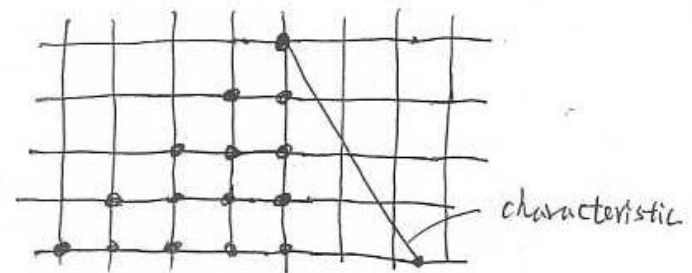


If $a > 0$, the right inequality is always true; the left inequality requires that $a\Delta t \leq \Delta x$, or $av \leq 1$.

Therefore, when $a > 0$, the CFL condition is $av \leq 1$.

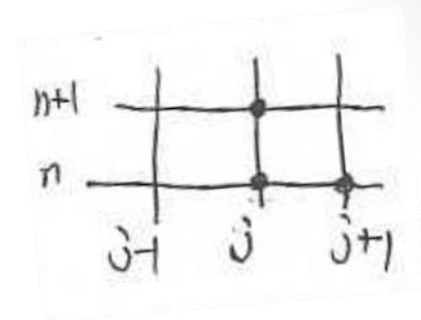
CFL condition $x_{j-n-1} \leq x_j - at_{n+1} \leq x_j$.

If $a < 0$, the right inequality is always false, which means that this scheme is always unstable.



For the numerical scheme

$$\frac{U_j^{n+1} - U_j^n}{\Delta t} + a \frac{U_{j+1}^n - U_j^n}{\Delta x} = 0 \quad (3)$$

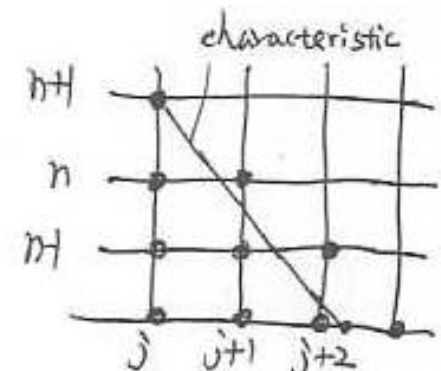


$$U_j^{n+1} = U_j^n - a\nu (U_{j+1}^n - U_j^n)$$

The domain of dependence of $u(x_j, t_{n+1})$ is $\{x_j - at_{n+1}\}$.

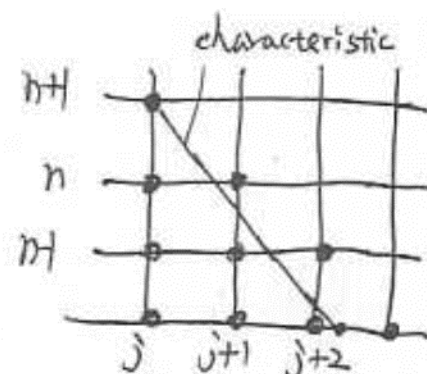
Numerical domain of dependence is $x_j \leq x \leq x_{j+n+1}$

CFL condition is $x_j \leq x_j - at_{n+1} \leq x_{j+n+1}$



If $a > 0$, the left inequality is always false, which means that this scheme is always unstable.

If $a < 0$, the CFL condition requires that $|a|\nu \leq 1$.



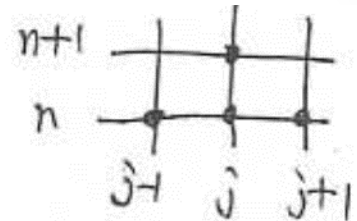
Therefore, when $a > 0$, we can use scheme (2), and when $a < 0$, we can use scheme (3). In both cases, the CFL condition is $|a|\nu \leq 1$.

This is the upwind scheme.

The CFL condition of the upwind scheme is the same as the stability condition obtained using Fourier analysis.

Now consider the numerical scheme

$$\frac{U_j^{n+1} - U_j^n}{\Delta t} + a \frac{U_{j+1}^n - U_{j-1}^n}{2\Delta x} = 0 \quad (1)$$

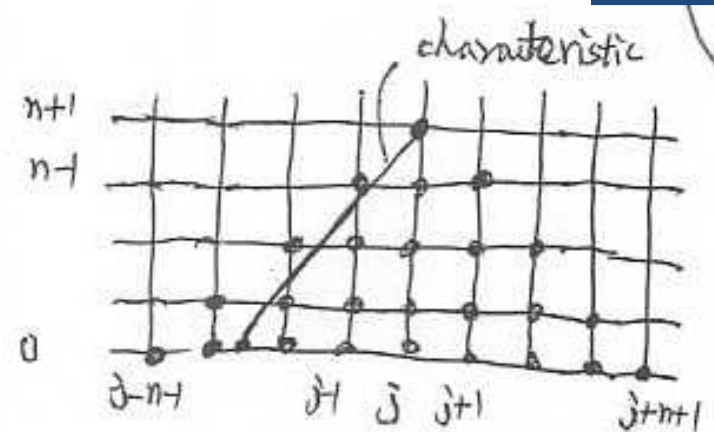


The domain of dependence of $u(x_j, t_{n+1})$

$$\{x_j - at_{n+1}\}.$$

Numerical domain of dependence

$$x_{j-n-1} \leq x \leq x_{j+n+1}$$



CFL condition $x_{j-n-1} \leq x_j - at_{n+1} \leq x_{j+n+1}$

or $|a|\nu \leq 1.$

The CFL condition is only a necessary condition. Using Fourier analysis, we know that this central difference scheme is always unstable.

2.7. Lax-Wendroff scheme

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$$

Perform Taylor expansion of $u(x_j, t_{n+1})$ at (x_j, t_n) .

$$u(x_j, t_{n+1}) = u(x_j, t_n) + \frac{\partial u}{\partial t}(x_j, t_n) \Delta t + \frac{1}{2} \frac{\partial^2 u}{\partial t^2}(x_j, t_n) (\Delta t)^2 + O((\Delta t)^3).$$

From the PDE,

$$\frac{\partial u}{\partial t} = -a \frac{\partial u}{\partial x}$$

$$\frac{\partial^2 u}{\partial t^2} = -a \frac{\partial^2 u}{\partial x \partial t} = -a \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial t} \right) = -a \frac{\partial}{\partial x} \left(-a \frac{\partial u}{\partial x} \right) = a^2 \frac{\partial^2 u}{\partial x^2}$$

Therefore,

$$\begin{aligned}u(x_j, t_{n+1}) &= u(x_j, t_n) - a \frac{\partial u}{\partial x}(x_j, t_n) \Delta t + \frac{a^2}{2} \frac{\partial^2 u}{\partial x^2}(x_j, t_n) (\Delta t)^2 + O((\Delta t)^3) \\&\approx u(x_j, t_n) - a \frac{\partial u}{\partial x}(x_j, t_n) \Delta t + \frac{a^2}{2} \frac{\partial^2 u}{\partial x^2}(x_j, t_n) (\Delta t)^2.\end{aligned}$$

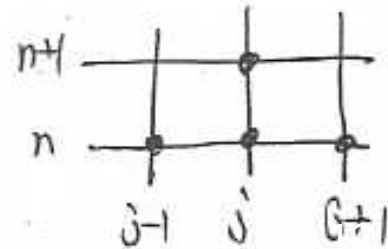
Using the approximation

$$\frac{\partial u}{\partial x}(x_j, t_n) \approx \frac{u(x_{j+1}, t_n) - u(x_{j-1}, t_n)}{2\Delta x},$$

$$\frac{\partial^2 u}{\partial x^2}(x_j, t_n) \approx \frac{u(x_{j+1}, t_n) - 2u(x_j, t_n) + u(x_{j-1}, t_n))}{(\Delta x)^2}$$

We have the Lax-Wendroff scheme

$$\nu = \frac{\Delta t}{\Delta x}$$



$$U_j^{n+1} = U_j^n - \frac{a\nu}{2}(U_{j+1}^n - U_{j-1}^n) + \frac{a^2\nu^2}{2}(U_{j+1}^n - 2U_j^n + U_{j-1}^n)$$

Truncation error

$$\begin{aligned} T(x_j, t_n) &= \frac{u(x_j, t_{n+1}) - u(x_j, t_n)}{\Delta t} + \frac{a\nu}{2\Delta t} [u(x_{j+1}, t_n) - u(x_{j-1}, t_n)] \\ &\quad - \frac{a^2\nu^2}{2\Delta t} [u(x_{j+1}, t_n) - 2u(x_j, t_n) + u(x_{j-1}, t_n)] \\ &= \left(\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} \right) + \frac{\Delta t}{2} \left(\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} \right) \\ &\quad + \frac{a}{6} \frac{\partial^4 u}{\partial x^4} (\Delta x)^2 + \frac{1}{6} \frac{\partial^3 u}{\partial t^3} (\Delta t)^2 + O((\Delta t)^3) + O((\Delta x)^3) \\ &= \frac{a}{6} \frac{\partial^3 u}{\partial x^3} (\Delta x)^2 + \frac{1}{6} \frac{\partial^3 u}{\partial t^3} (\Delta t)^2 + O((\Delta t)^3) + O((\Delta x)^3). \end{aligned}$$

Second order in both t and x .

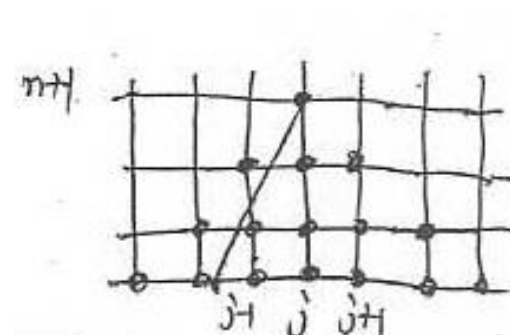
CFL condition

The domain of dependence of $u(x_j, t_{n+1})$ is $\{x_j - at_{n+1}\}$.

Numerical domain of dependence is $x_{j-n-1} \leq x \leq x_{j+n+1}$

CFL condition is $x_{j-n-1} \leq x_j - at_{n+1} \leq x_{j+n+1}$

$$|a|\nu \leq 1.$$



Fourier analysis of stability

$$U_j^n = [\lambda(k)]^n e^{ikx_j}$$

$$\lambda(k) = 1 - 2a^2\nu^2 \sin^2 \frac{k\Delta x}{2} - ia\nu \sin k\Delta x$$

$$|\lambda(k)|^2 = 1 - 4a^2\nu^2(1 - a^2\nu^2) \sin^4 \frac{k\Delta x}{2}$$

This scheme is stable, i.e., $|\lambda(k)|^2 \leq 1$, if and only if

$$|a|\nu \leq 1.$$

An example

$$\frac{\partial u}{\partial t} + a(x, t) \frac{\partial u}{\partial x} = 0, \quad x \geq 0, \quad t \geq 0$$

where

$$a(x, t) = \frac{1 + x^2}{1 + 2xt + 2x^2 + x^4}$$

The initial condition is

$$u(x, 0) = u_0(x) = \begin{cases} 1 & \text{if } 0.2 \leq x \leq 0.4, \\ 0 & \text{otherwise,} \end{cases}$$

The boundary condition is

$$u(0, t) = 0.$$

The exact solution is

$$u(x, t) = u_0 \left(x - \frac{t}{1 + x^2} \right)$$

Using the upwind scheme (here $a > 0$)

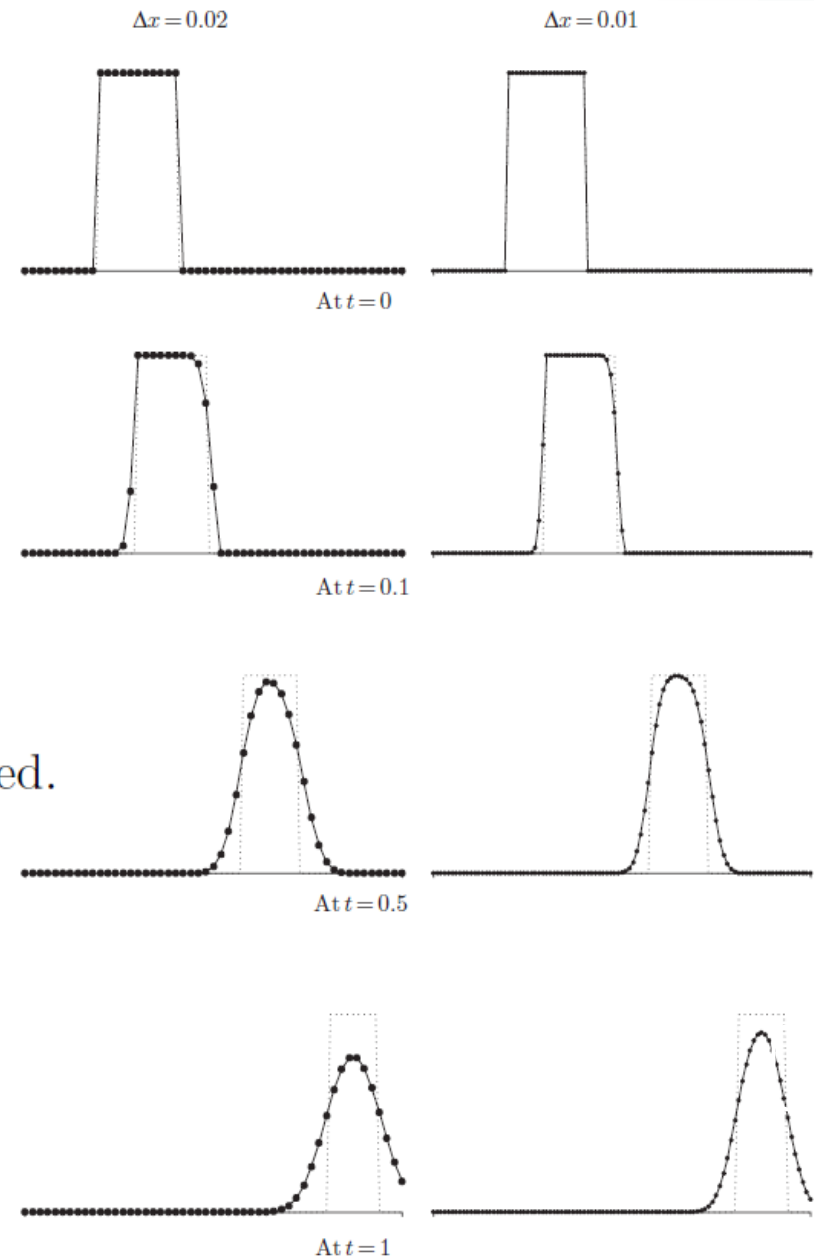
$$\frac{U_j^{n+1} - U_j^n}{\Delta t} + a(x_j, t_n) \frac{U_j^n - U_{j-1}^n}{\Delta x} = 0$$

$$\Delta t = \Delta x$$

Since $a(x, t) \leq 1$

the CFL stability condition is satisfied.

Significant smoothing of the edges of the pulse compared with the exact solution.

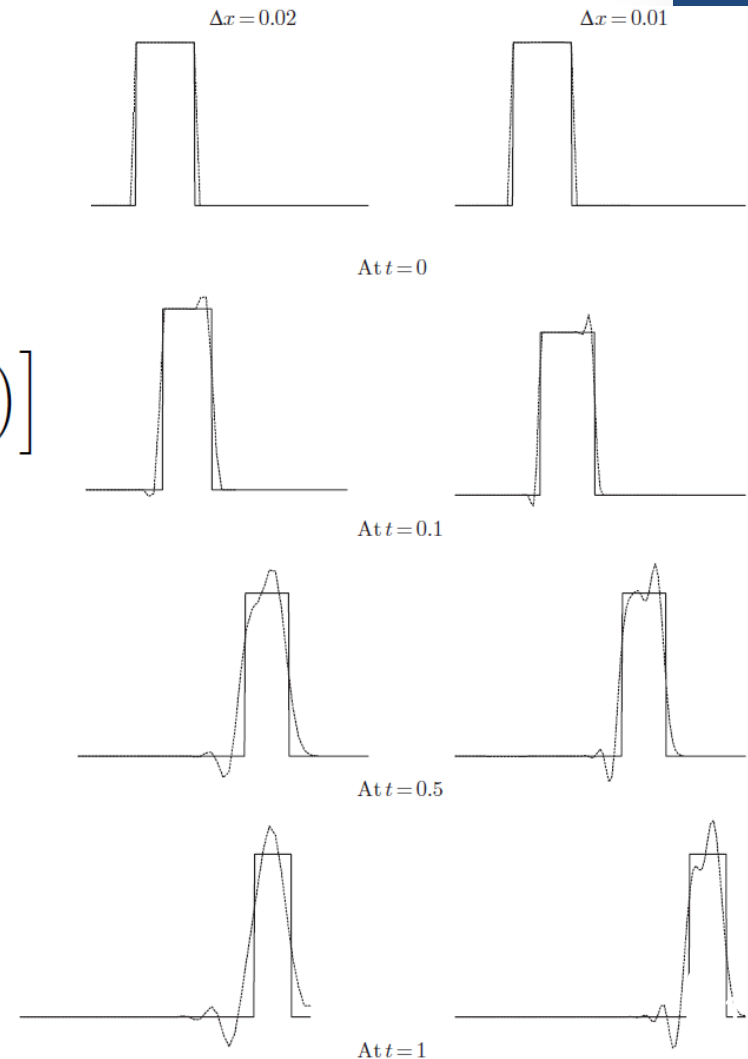


Using the Lax-Wendroff scheme

$$\begin{aligned}
 U_j^{n+1} = & U_j^n - \frac{a_j^n \nu}{2} (U_{j+1}^n - U_{j-1}^n) \\
 & + \frac{1}{2} (\Delta t)^2 \left[- \left(\frac{\partial a}{\partial t} \right)_j^n \frac{U_{j+1}^n - U_{j-1}^n}{2\Delta x} \right. \\
 & \left. + \frac{a_j^n}{(\Delta x)^2} \left(a_{j+\frac{1}{2}}^n (U_{j+1}^n - U_j^n) - a_{j-\frac{1}{2}}^n (U_j^n - U_{j-1}^n) \right) \right]
 \end{aligned}$$

$$\Delta t = \Delta x$$

- Maintains the height and width of the pulse better than the upwind scheme.
- Generates oscillations



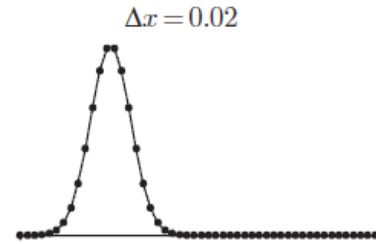
Smooth initial condition

$$u(x, 0) = \exp[-10(4x - 1)^2].$$

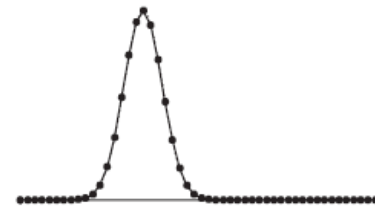
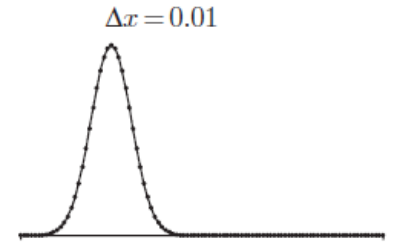
Using the Lax-Wendroff scheme

$$\Delta t = \Delta x$$

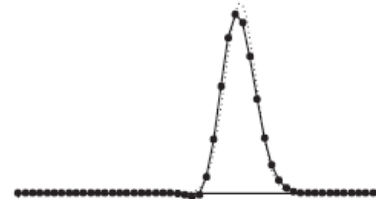
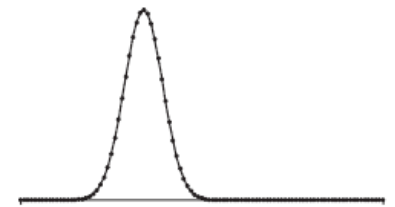
Considerably more accurate



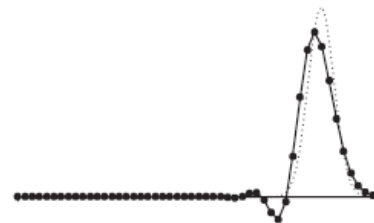
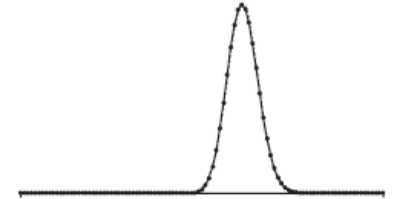
At $t=0$



At $t=0.1$



At $t=0.5$



At $t=1$

